Geometric Flow Theory (TGF v6.5)

A Covariant Effective Viscous-Scalar Extension of ΛCDM

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ABSTRACT

We present the Geometric Flow Theory (TGF v6.5), a covariant effective field theory extending General Relativity by coupling spacetime dynamics to a relativistic imperfect fluid and a scalar field ϕ\phiϕ. In this framework, cosmic acceleration and dark sector phenomena emerge from viscous transport processes between underdense (void-like) and overdense (sink-like) regions.

Dark matter and dark energy are replaced by effective shear and bulk viscosity terms μ(ϕ)\mu(\phi)μ(ϕ) and ζ(ϕ)\zeta(\phi)ζ(ϕ), which encode environment-dependent deviations from ΛCDM. The theory predicts late-time deviations in the Hubble expansion and a unique observable: a non-zero correlation between luminosity distance residuals and cosmic void structure.

We show that the model reduces to ΛCDM in the limit η→0\eta \to 0η→0, where η\etaη is a single effective coupling parameter controlling the strength of structure–distance coupling.

1. INTRODUCTION

The ΛCDM model successfully describes cosmological observations but relies on two unknown components: dark matter and dark energy. TGF proposes an alternative interpretation in which these phenomena emerge from dissipative relativistic flow dynamics of spacetime itself.

We model the Universe as a covariant viscous medium governed by General Relativity coupled to a scalar field representing geometric flow structure.

2. THEORETICAL FRAMEWORK

2.1 Action

S=∫d4x−g[116πG(R−2Λ)−12(∇ϕ)2−V(ϕ)+Lfluid+Lvisc+αϕR]S = \int d^4x \sqrt{-g} \left[\frac{1}{16\pi G}(R - 2\Lambda) -\frac{1}{2}(\nabla \phi)^2 - V(\phi) + \mathcal{L}_{fluid} + \mathcal{L}_{visc} + \alpha \phi R \right]S=∫d4x−g[16πG1(R−2Λ)−21(∇ϕ)2−V(ϕ)+Lfluid +Lvisc+αϕR]

2.2 Effective stress-energy tensor

Tμνeff=(ρ+P)uμuν+Pgμν−2μ(ϕ)σμν−ζ(ϕ)θgμνT_{\mu\nu}^{eff} = (\rho + P)u_\mu u_\nu + P g_{\mu\nu} -2\mu(\phi)\sigma_{\mu\nu} -\zeta(\phi)\theta g_{\mu\nu}Tμνeff=(ρ+P)uμuν+Pgμν −2μ(ϕ)σμν−ζ(ϕ)θgμν

where:

- σμν\sigma_{\mu\nu}σμν: shear tensor
- θ=∇ μuμ\theta = \nabla_\mu u^\muθ=∇ μuμ: expansion scalar
- μ(ϕ),ζ(ϕ)\mu(\phi), \zeta(\phi)μ(ϕ),ζ(ϕ): viscosity functions

2.3 Scalar field equation

□ϕ−dVdϕ+αR+β∇ μuμ=0\square \phi - \frac{dV}{d\phi} + \alpha R + \beta \nabla_\mu u^\mu = 0□ϕ−dϕdV+αR+β∇ μuμ=0

3. COSMOLOGICAL BACKGROUND

3.1 Modified Friedmann equation

H2(z)=8πG3[ρm+ρϕ−3Hζ(ϕ)]H^2(z) = \frac{8\pi G}{3} \left[\rho_m + \rho_\phi - 3H\zeta(\phi) \right]H2(z)=38πG[ρm+ρϕ−3Hζ(ϕ)]

The viscous term introduces an effective late-time deviation from ΛCDM.

3.2 Minimal parameter reduction

We define a single observable coupling parameter:

η\etaη

such that:

H(z)=HΛCDM(z) [1+ηg(z)]H(z) = H_{\Lambda CDM}(z)\,[1 + \eta g(z)]H(z)=HΛCDM (z)[1+ηg(z)]

where g(z)g(z)g(z) is a known redshift-dependent structure function.

4. OBSERVATIONAL SIGNATURE

4.1 Primary observable (key prediction)

We define:

K=⟨ δDL(z,n^)⋅ δvoid(n^)⟩ \mathcal{K} = \left\langle \delta D_L(z,\hat{n}) \cdot \delta_{void}(\hat{n}) \right\rangleK=⟨ δDL(z,n^)⋅ δvoid(n^)⟩

4.2 ΛCDM prediction

KΛCDM=0\mathcal{K}_{\Lambda CDM} = 0KΛCDM=0

4.3 TGF prediction

KTGF=ηF(z)\mathcal{K}_{TGF} = \eta F(z)KTGF=ηF(z)

with:

- η≠0⇒\eta \neq 0 \Rightarrowη□=0⇒ structured correlation
- sign and amplitude depend on void gradients

5. ORDER-OF-MAGNITUDE ESTIMATE

ΔDLDL∼10−4−10−3\frac{\Delta D_L}{D_L} \sim 10^{-4} - 10^{-3}DLΔDL∼10−4−10−3

Signal-to-noise for upcoming surveys:

- single object: SNR∼10−2SNR \sim 10^{-2}SNR∼10−2
- full survey: SNR∼3−30SNR \sim 3 - 30SNR∼3−30

6. STATISTICAL FRAMEWORK

Likelihood comparison:

LTGF=LSNLBAOLCMBLLSS\mathcal{L}_{TGF} = \mathcal{L}_{SN} \mathcal{L}_{BAO} \mathcal{L}_{CMB} \mathcal{L}_{LSS}LTGF=LSNLBAOLCMBLLSS

Bayesian evidence is controlled by:

η\etaη

with ΛCDM recovered in the limit η→0\eta \to 0η→0.

7. FALSIFIABILITY CRITERION

TGF is falsified if:

1. K=0\mathcal{K} = 0K=0 within observational uncertainty
2. no correlation between SN Ia residuals and void structure
3. no environment dependence of cosmological distances

8. DISCUSSION

TGF represents an effective dissipative cosmology in which dark sector effects emerge from viscous spacetime flow. The model is consistent with current constraints provided $|\eta| \lesssim 10^{-3}$, and predicts small but measurable deviations from ΛCDM at late times.

Its key distinguishing feature is not background expansion, but **correlated anisotropic structure in cosmological distance measurements**.

9. CONCLUSION

We have presented a covariant viscous scalar-field cosmology reducing to a single effective parameter η. The theory is fully falsifiable and predicts a unique observational signature: a non-zero correlation between cosmological distance residuals and cosmic void distribution.

Future surveys such as Euclid and DESI will be able to test this prediction directly.